

Cellular Respiration — Complete Notes

Grade 11

Life Sciences

CAPS

What you'll learn

- What cellular respiration is and why every living cell needs it
- The structure of ATP and why it is the cell's energy currency
- The balanced equation for aerobic respiration
- The three stages of aerobic respiration and where each happens
- Anaerobic respiration in muscle cells and in yeast
- How to compare photosynthesis and respiration
- How to answer process, comparison and data questions

What is cellular respiration?

Cellular respiration is the controlled release of energy from **glucose** (and other organic molecules) to make **ATP**, the cell's energy currency. It happens in **every living cell**, all the time — a plant respire even while it is photosynthesising.

- It is a **catabolic** process (it breaks larger molecules into smaller ones) and releases energy.
- The energy released is not wasted as heat; it is captured in **ATP**.
- **Glucose** is the main substrate, but fats and proteins can also be respired.

ATP — the energy currency

ATP stands for **adenosine triphosphate**. It is made of adenine, ribose and **three phosphate groups**.



ATP = adenosine + 3 phosphates

- ATP stores energy in the bonds between its phosphate groups.
- When the **last phosphate** is removed, ATP becomes **ADP + Pi** and **energy is released** for cell activities (muscle contraction, active transport, cell division, building molecules).
- Respiration **recharges** ADP back into ATP.

Aerobic respiration

Aerobic respiration needs **oxygen**. It releases much more energy than anaerobic respiration.

Word equation:

glucose + oxygen → carbon dioxide + water + energy (ATP)

Balanced chemical equation:

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$

The three stages

Stage	Where it happens	Main product
1. Glycolysis	Cytoplasm	2 pyruvate (from 1 glucose)
2. Krebs cycle	Mitochondrial matrix	CO ₂ , NADH, FADH ₂ , a little ATP
3. Oxidative phosphorylation	Inner mitochondrial membrane (cristae)	Lots of ATP, water

Stage 1 — Glycolysis (cytoplasm)

- **Glucose (6C)** is split into **2 pyruvate molecules (3C)**.
- No oxygen is needed for this stage.
- **2 ATP** are used and **4 ATP** are produced, so there is a **net gain of 2 ATP**, plus **2 NADH**.
- This is the only stage common to **aerobic and anaerobic** respiration.

Stage 2 — Krebs cycle (mitochondrial matrix)

- Pyruvate moves into the **mitochondria** and is converted to **acetyl-CoA**, releasing **CO₂**.
- Acetyl-CoA enters the **Krebs cycle** in the **matrix**.
- More **CO₂** is released, and small amounts of **ATP**, **NADH** and **FADH₂** are produced.

Stage 3 — Oxidative phosphorylation (inner membrane)

- **NADH** and **FADH₂** give up their electrons to the **electron transport chain** in the **cristae**.
- The released energy pumps H⁺ ions and is used by **ATP synthase** to make **ATP**.
- **Oxygen is the final electron acceptor**, combining with H⁺ to form **water (H₂O)**.
- This stage produces **most of the ATP** (about 34 molecules).

ATP total

Per glucose molecule, aerobic respiration yields about **38 ATP** in total (2 from glycolysis + 2 from the Krebs cycle + about 34 from oxidative phosphorylation). Note that some textbooks quote a lower total (30–32) because energy is used to move molecules into the mitochondria — for CAPS, learn the **≈ 38** figure but understand that it is an approximation.

Anaerobic respiration (without oxygen)

When oxygen is in short supply, cells carry out **anaerobic respiration**. It releases much **less** energy — only **2 ATP** per glucose.

Lactic acid fermentation — muscle cells

- Glucose → **2 lactic acid** + energy (2 ATP).

- Happens in the **cytoplasm** of muscle cells during intense exercise, when oxygen cannot reach the muscles fast enough.
- The **lactic acid** builds up, causing muscle fatigue, cramp and an oxygen debt.
- Later, oxygen is used to break down the lactic acid (mostly in the liver), which is why you keep breathing heavily after exercise.

Alcoholic fermentation — yeast

- Glucose → **2 ethanol (alcohol) + 2 CO₂ + energy (2 ATP)**.
- Used commercially: **yeast** is used in **brewing** (ethanol in beer and wine) and **baking** (CO₂ makes bread rise).

Comparison

Feature	Aerobic	Anaerobic
Oxygen	Needed	Not needed
Location	Cytoplasm + mitochondria	Cytoplasm only
ATP per glucose	≈ 38	2
Products in muscle	CO ₂ + H ₂ O	Lactic acid
Products in yeast	CO ₂ + H ₂ O	Ethanol + CO ₂

Comparing photosynthesis and respiration

Feature	Photosynthesis	Respiration
Energy	Stores light energy as food	Releases energy as ATP
Gas used	Carbon dioxide	Oxygen (aerobic)
Gas released	Oxygen	Carbon dioxide
Occurs in	Chloroplasts (green cells)	All living cells
When	Only in light (light phase)	All the time
Food	Glucose is made	Glucose is broken down
Enzymes	Yes	Yes

They are **opposite** reactions, but neither is simply the reverse of the other.

Factors affecting the rate of respiration

- **Temperature:** the rate rises to an optimum then falls as the enzymes denature.
- **Oxygen availability:** without oxygen, only anaerobic respiration occurs, giving far less ATP.
- **Glucose availability:** the substrate must be present.
- **Cell activity and energy demand:** more active cells respire faster.

Worked examples

Worked example 1 — Explain where and why

Explain why a sprinter's muscle cells produce lactic acid during a race, but not at rest.

Step 1 — At rest, enough O₂ reaches the muscles, so **aerobic respiration** meets the energy demand.

Step 2 — During a sprint the demand for ATP is very high and the heart and lungs cannot deliver O₂ fast enough.

Step 3 — Without enough O₂, the muscles switch to **anaerobic respiration**.

Step 4 — Anaerobic respiration in muscle converts glucose to **lactic acid**, with only 2 ATP per glucose.

Step 5 — Lactic acid builds up, causing fatigue and cramp, and creates an **oxygen debt** that is repaid after the race.

Answer: Because oxygen delivery cannot keep up with demand, the muscles respire anaerobically, producing lactic acid.

Worked example 2 — Compare and calculate

Compare the energy yield: how many times more ATP does aerobic respiration produce than anaerobic (using 38 vs 2)?

Step 1 — Aerobic ≈ 38 ATP; anaerobic = 2 ATP.

Step 2 — Ratio = $38 \div 2 = 19$.

Answer: about **19 times** more ATP per glucose.

Worked example 3 — Interpret data

A sealed flask of yeast and glucose shows a rise in CO₂ and the smell of alcohol after a few hours, with the yeast still respiring. Explain.

Step 1 — The flask is sealed, so oxygen runs out, creating anaerobic conditions.

Step 2 — Yeast carries out **alcoholic fermentation**: glucose $\rightarrow$ ethanol + CO₂.

Step 3 — The CO₂ causes the rise in gas, and the ethanol produces the alcoholic smell.

Answer: The yeast respire anaerobically, producing ethanol and CO₂.

Common mistakes

- Saying respiration means "breathing". **Breathing** (gaseous exchange) brings in oxygen; **cellular respiration** releases energy in cells.
- Writing that anaerobic respiration makes **no** energy. It makes **2 ATP** (just much less).
- Thinking plants respire only at night. Plants respire **all the time**; they photosynthesise only in light.
- Putting glycolysis in the mitochondria — glycolysis happens in the **cytoplasm**.
- Mixing up the fermentation products: **muscle** $\rightarrow$ **lactic acid**; **yeast** $\rightarrow$ **ethanol + CO₂**.
- Forgetting that **oxygen is the final electron acceptor** in aerobic respiration, forming water.
- Confusing ATP and ADP: ATP has **three** phosphates; ADP has **two**.

Quick summary / cheat sheet

- **Aerobic:** $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2 \rightarrow 6\text{CO}_2 + 6\text{H}_2\text{O} + \text{energy} (\approx 38 \text{ ATP})$.
- **Stages:** glycolysis (cytoplasm) $\rightarrow$ Krebs cycle (matrix) $\rightarrow$ oxidative phosphorylation (cristae).

- **Anaerobic:** muscle → lactic acid (2 ATP); yeast → ethanol + CO₂ (2 ATP).
- **ATP** = energy currency; made from ADP + Pi.
- **Oxygen** is the final electron acceptor (forms water).
- Photosynthesis stores energy; respiration releases it.

Practice questions

1. Write the word and balanced chemical equations for aerobic respiration.
2. Name the three stages of aerobic respiration and state where each occurs.
3. How many ATP molecules are produced in total during aerobic respiration, and how many during anaerobic respiration?
4. Give the products of anaerobic respiration in (a) human muscle cells and (b) yeast.
5. Explain why you continue to breathe heavily for a while after sprinting.
6. State two differences between aerobic and anaerobic respiration.
7. If aerobic respiration produces about 38 ATP and anaerobic respiration produces 2 ATP per glucose, calculate how many ATP 5 glucose molecules would produce aerobically.

Answers

1. Words: glucose + oxygen → carbon dioxide + water + energy. Symbols: $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O + \text{energy}$.
2. Glycolysis → cytoplasm; Krebs cycle → mitochondrial matrix; oxidative phosphorylation → inner mitochondrial membrane (cristae).
3. Aerobic ≈ 38 ATP; anaerobic = 2 ATP.
4. (a) Lactic acid; (b) ethanol + carbon dioxide.
5. The muscles built up an **oxygen debt**; extra oxygen is needed to break down the accumulated lactic acid (in the liver).
6. Any two: aerobic needs oxygen, anaerobic does not; aerobic ≈ 38 ATP, anaerobic 2 ATP; aerobic occurs in the cytoplasm and mitochondria, anaerobic only in the cytoplasm; aerobic products are CO₂ + H₂O, anaerobic products are lactic acid or ethanol + CO₂.
7. $38 \times 5 = \mathbf{190 \text{ ATP}}$.

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